LAB 4: SIMULATION OF ANGULAR VELOCITY CONTROL FOR A DC MOTOR

```
%--------------------------------------
% Fill in your student Name and ID.
Students.Name = 'Seyed Yazdan Zinati';
Students.ID = '260928077';
%--------------------------------------
```

0. Objectives

In this lab, we will run different simulation experiments to track a desired reference signal with a simulated DC Motor. We will first run an open-loop experiment, then we will use state feedback to control the angular velocity of DC motor.

1. A State-Space Representation of DC Motor

A DC motor is a device that converts armature current into mechanical torque. A schematic of a DC motor is shown in the following

<<fig_dcmotorshematic.png>>

We can model the DC motor using the armature current i and the angular speed ω. The dynamics of the DC motor are given by the following two coupled differential equations:

$$J\frac{d\omega}{dt} + b\omega = K_t i$$

$$L\frac{di}{dt} + Ri = v - K_e\omega$$

where the parameters are defined as follow

```
Param.Kt = 0.034; % Torque proportionality constant (Kg m2 /s2/A) (N.m/A)
Param.Ke = 0.034; % Back electromotive proportionality constant (V/rad/s)
Param.L = 0.018; % Electric inductance (H)
Param.R = 8.4; % Terminal resistance (Ohm)
Param.J = 2e-05; % Rotor moment of inertia (kg.m2)
Param.b = 1e-06; % Rotor viscous friction constant (Kg.m2/s) (N.m.s)
```

From these differential equations we can represent the DC Motor model by the following state-space representation

$$\frac{d}{dt}x(t) = \begin{pmatrix} -\frac{b}{J} & \frac{K_t}{J} \\ -\frac{K_e}{L} & -\frac{R}{L} \end{pmatrix} x(t) + \begin{pmatrix} 0 \\ \frac{1}{L} \end{pmatrix} v(t)$$

where $v(t)$ is the voltage and the state vector contains the angular velocity $\omega(t)$ and the current $i(t)$

$$x(t) = \begin{pmatrix} \omega(t) \\ i(t) \end{pmatrix}$$

Question 1 (1 mark)

Write the state-space representation of the DC Motor (Only A and B matrices).

```
A = [-Param.b/Param.J, Param.Kt/Param.J; -Param.Ke/Param.L, -Param.R/Param.L]
```

```
A = 2×2
10^3 ×
   -0.0000    1.7000
   -0.0019   -0.4667
```

```
B = [0; 1/Param.L]
```

```
B = 2×1
         0
   55.5556
```

2. Desired specifications

The objective of this lab is to track a reference signal ω_{ref} while satisfying the following requirements:

- Overshoot less than 10%.
- 5%-settling time less than 0.2 seconds.
- Steady state gain of 1 $\pm$ 5%.

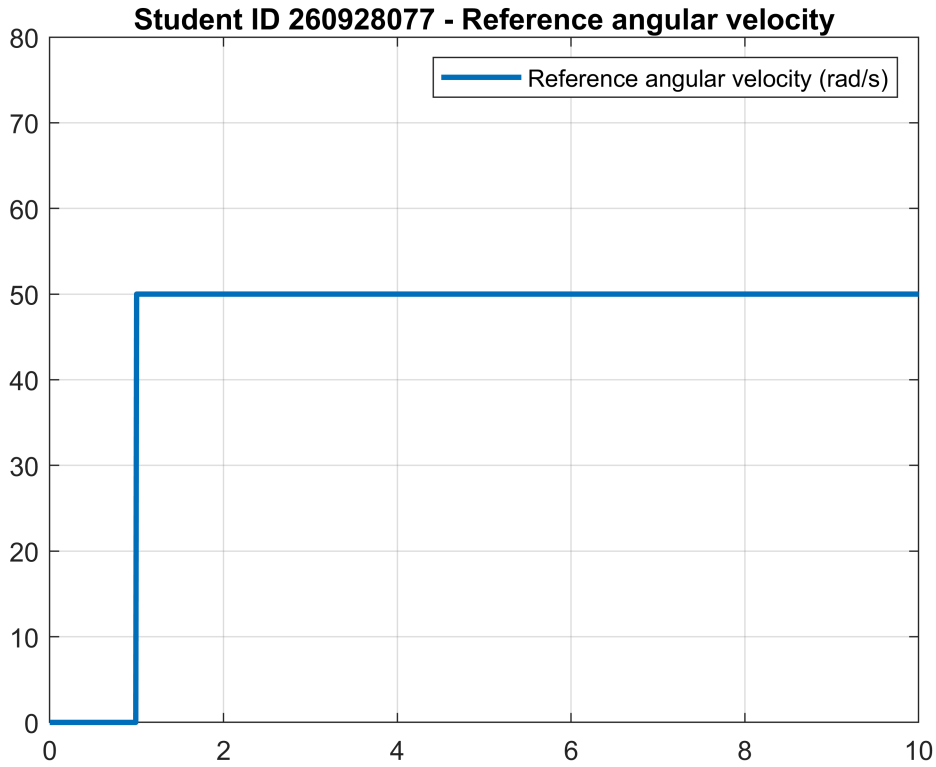
The Reference angular velocity is defined by the following:

$$\omega_{ref}(t) = \begin{cases} 0, & 0 \leq t < 1 \\ 50, & t \geq 1 \end{cases}$$

In **MATLAB** we define ω_{ref} by:

```
T = 10; % Simulation duration
dt = 0.01; % Simulation step time
t = (0:dt:T)'; % Time array
TStart = 1; % Step start
wRef = zeros(size(t));
wRef(t >= TStart) = 50; % rad/s

figure;
clf;
plot(t, wRef, 'LineWidth', 2.0)
legend('Reference angular velocity (rad/s)');
xlim([0, T])
ylim([0, 80])
title(sprintf('Student ID %s - Reference angular velocity', Students.ID));
grid on
```



3. Open-loop Simulation Experiment

The state-space open-loop system is presented below.

<<fig_lab04_openloopdiagram.png>>

Note that if we think of the angular velocity $\omega(t)$ as the output of this system, then $C = \begin{pmatrix} 1 & 0 \end{pmatrix}$ and

$$\omega(t) = Cx(t)$$

In this case, the transfer function from the voltage $v(t)$ to the angular velocity $\omega(t)$ is given by

$$G_o(s) = C(sI - A)^{-1}B$$

In **MATLAB**, we can find the transfer function of a state-space model using the function **ss** and **tf**, as shown below.

```
C = [1, 0];
D = 0;
Go = ss(A, B, C, D)
```

```
Go =

  A =
         x1      x2
   x1   -0.05    1700
   x2  -1.889  -466.7

  B =
```

```
         u1
   x1     0
   x2  55.56

  C =
       x1  x2
   y1   1   0

  D =
       u1
   y1   0
```

```
Continuous-time state-space model.
```

```
tf(Go)
```

```
ans =

        9.444e04
  --------------------
  s^2 + 466.7 s + 3234

Continuous-time transfer function.
```

Question 2 (1 mark)

Recall that the DC gain can be obtained from a Laplace transfer function, by setting the laplace variable s to zero. From the state-space model representation (A, B, C, D), calculate the DC gain.

```
% dcGainOL = 9.444e04/3234
dcGainOL = dcgain(Go)
```

```
dcGainOL = 29.1996
```

Observe that if the open-loop system is a stable system. The DC Motor will be responded to a reference step signal without using a controller. However, the response will be determined by the eigenvalues of the open-loop A matrix (or equivalently, the poles of the open-loop transfer function).

Question 3 (1 mark)

Determine the poles of the open-loop transfer function from the state-space matrix A (Hint: **`help eig`**). Is the system stable? (Justify your answer.)

```
eigenvalues = eig(A);

p1 = eigenvalues(1)
```

```
p1 = -7.0363
```

```
p2 = eigenvalues(2)
```

```
p2 = -459.6804
```

```
% We can observe that the eigenvalues of A (or equivalently the poles of
% the open-loop transfer funtion) are both negative and therefore situated
```

```
% in the LHP. So the system is stable.
```

Following a fixed input v_0 the angular velocity ω will attain a steady angular velocity $\omega_\infty = DC_{gain} v_0$.

We can simulate the response of an LTI system to an arbitrary input using the **`lsim`** function.

Question 4 (0.5 mark)

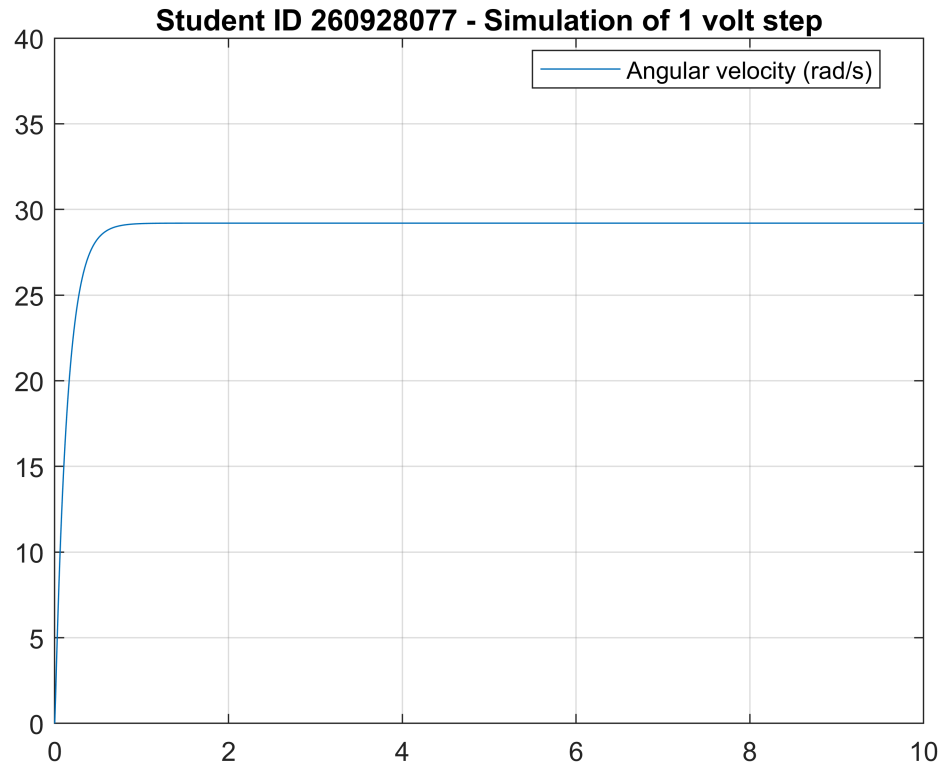
Complete the following code to simulate the response of the system to a 1 volt input signal, and plot the results.

```
v = ones(size(t)); % create a signal with one volt.
w = lsim(Go, v, t);

% Plot results
figure;
clf;

% plot the angular velocity w
plot(t, w)

legend('Angular velocity (rad/s)', ...
    'location', 'best');
xlim([0, T])
ylim([0, 40])
title(sprintf('Student ID %s - Simulation of 1 volt step', Students.ID));
grid on
```



In open-loop we can generate the voltage v from ω_{ref} through the gain N_o.

$$v(t) = N_o \omega_{ref}(t)$$

Question 5 (1 mark)

Find the value of N_o such the final value of the angular velocity output ω is the same as ω_{ref} (i.e. 50 rad/s)

```
% No = 0.03425 (Found using an experimental method)

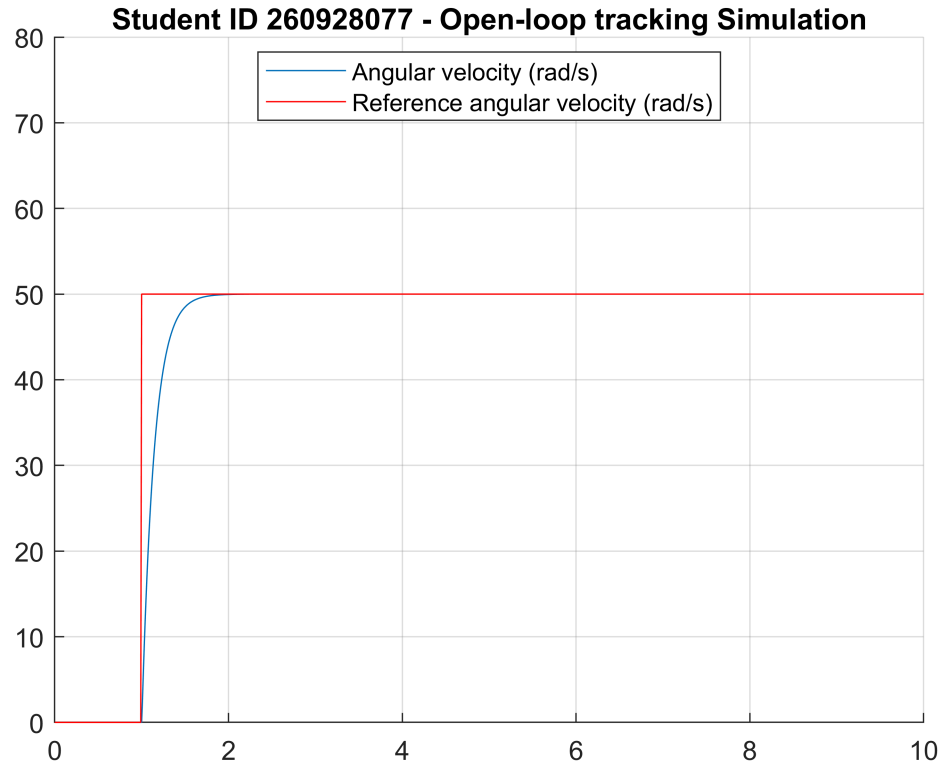
No = 1/dcGainOL
```

```
No = 0.0342
```

```
% The voltage is generated from wRef and No
v = No * wRef;
% Simulate the output angular velocity using lsim
w = lsim(Go, v, t);


% Plot results
figure;
clf;
hold on
% Superpose the angular velocity w and the reference wRef
plot(t, w)
plot(t, wRef, 'r')

legend('Angular velocity (rad/s)', ...
    'Reference angular velocity (rad/s)', ...
    'location', 'best');
xlim([0, T])
ylim([0, 80])
title(sprintf('Student ID %s - Open-loop tracking Simulation', Students.ID));
grid on
```



Question 6 (0.5 marks)

Based on the plot above, determine if all the requirements are satisfied. Justify your answer.

```
stepinfo(w, "SettlingTimeThreshold", 0.05)
```

```
ans = struct with fields:
        RiseTime: 31.2225
    SettlingTime: 143.8004
     SettlingMin: 45.0198
     SettlingMax: 50.0000
       Overshoot: 0
      Undershoot: 0
            Peak: 50.0000
        PeakTime: 593
```

```
finalW = mean(w(t>=6));
finalWred = mean(wRef(t>=6));
SSgain = finalW/finalWred
```

```
SSgain = 1.0000
```

```
% We had the following requirements defined in 2. :
% 1 - Overshoot less than 10%
```

```
% 2 - 5%-settling time less than 0.2 seconds
% 3 - Steady state gain of 1±5%

% We can see from the above plot that the OS% is 0 so OS% requirements are
% met. The 5% settling time is about 0.438 s which is larger than the 0.2 s
% to satisfy our requirements. So the settling time doesn't meet our
% requirements. The SS gain is also very close to 1 and withing our defined
% error bound of 5%.

% So as the requirements for Ts are not met, not all
% the requirements are satisfied based on the above plot.
```

4. Closed-Loop Simulation Experiment

The closed-loop system is presented in the following figure.

<<fig_lab04_clsoedloopdiagram1.png>>

The full-state feedback control law is defined as follow

$$v(t) = N_c \omega_{ref} - Kx(t)$$

where gain K is a 1 by 2 matrix and the gain N_c is forward gain that will ensure achieving ω_{ref} at steady state.

Actually, this diagram is equivalent to an open-loop diagram but with a different state transition matrix $A_{cl} = A - BK$

<<fig_lab04_clsoedloopdiagram2.png>>

Doing a state feedback with gain K is equivalent to changing the system dynamics from A to $A - BK$. Recall that the eigenvalues of a state transition matrix A are the poles of the system.

We want to choose K to specify the new poles of the closed-loop system $A - BK$ in a way that satisfies the desired specifications

- Overshoot less than 10%.
- 5%-settling time less than 0.2 seconds.

Then we will choose N_c similarly to the open-loop section to satisfy the steady state requirements:

- Steady state gain of 1 $\pm$ 5%.

Question 7 (2 marks)

Calculate the desired poles that satisfies the specifications (Use formula sheet for transient response of 2nd order system in myCourses if needed):

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

T_s (settling time) $\frac{4.6}{\sigma}$ (1% criterion)

$\frac{4}{\sigma}$ (2% criterion)

$\frac{3}{\sigma}$ (5% criterion)

```
% we use the 2 above expressions to find zeta and wn to subsitute in
% Poles = -zeta*wn (+/-) wn*sqrt(zeta^2 - 1)


zeta = -(log(10/100))/sqrt(pi.^2 + (log(10/100)).^2)
```

```
zeta = 0.5912
```

```

% 3/(wn*zeta) = 0.2
wn = 3/(zeta*0.2)
```

```
wn = 25.3741
```

```


p1d = -zeta*wn + wn * sqrt(zeta^2 - 1)
```

```
p1d = -15.0000 + 20.4656i
```

```
p2d = -zeta*wn - wn * sqrt(zeta^2 - 1)
```

```
p2d = -15.0000 - 20.4656i
```

Instead of manual calculation of the value of the gain K that place the poles of $A - BK$ at [p1d, p2d], we will use the **`place`** command to calculate the gain K

```
K = place(A, B, [p1d, p2d]);
```

Question 8 (1 mark)

The closed-loop transfer function is

$$G_c(s) = C(sI - (A - BK))^{-1}B$$

1. What is the value of the DC gain of the closed-loop system.
2. What is the value N_c needed to compensate for the DC gain of the closed-loop system.

```
C = [1, 0];
D = 0;
Gc = ss((A-B*K), B, C, D)
```

```
Gc =

  A =
           x1       x2
   x1    -0.05     1700
   x2  -0.3779   -29.95

  B =
         u1
   x1     0
   x2  55.56

  C =
       x1  x2
   y1   1   0

  D =
       u1
   y1   0

Continuous-time state-space model.
```

```
tf(Gc)
```

```
ans =

       9.444e04
  ------------------
  s^2 + 30 s + 643.8

Continuous-time transfer function.
```

```
% dcGainCL = 9.444e04/643.8
dcGainCL = dcgain(Gc)
```

```
dcGainCL = 146.6887
```

```
Nc = 1/dcGainCL
```

```
Nc = 0.0068
```

Question 9 (1 mark)

Complete the following to simulate angular velocity control with state feedback that tracks the desired ω_{ref}.

```
% Define the closed-loop system using ss :
```

```
% Gc = ss((A-B*K), B, C, D) (Already defined above)


% Simulate the output angular velocity using lsim
w = lsim(Gc, Nc*wRef, t);


% We also simulate the voltage for visualization
v = -lsim(ss(Gc.A, Gc.B, eye(2), Gc.D), Nc*wRef, t) * K' + Nc * wRef;

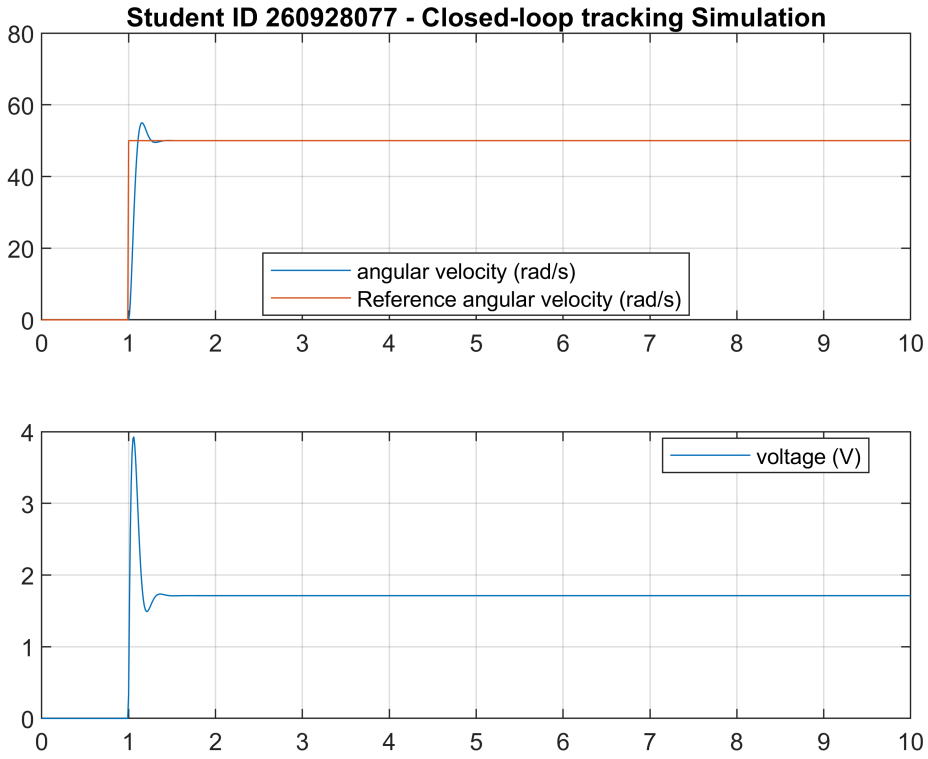

figure;
clf;
subplot(2, 1, 1)

% Superpose angular velocity w with the angular velocity reference
plot(t, w)
hold on
plot(t, wRef)
hold off

legend('angular velocity (rad/s)', ...
    'Reference angular velocity (rad/s)', ...
    'location', 'best');
xlim([0, T])
ylim([0, 80])
grid on
title(sprintf('Student ID %s - Closed-loop tracking Simulation', Students.ID));

subplot(2, 1, 2)
% Plot the voltage
plot(t, v)

legend('voltage (V)', 'location', 'best');
xlim([0, T])
grid on
```



Question 10 (1 marks)

Based on the plot above, determine if all the requirements are satisfied. Justify your answer.

```
stepinfo(w, "SettlingTimeThreshold", 0.05)
```

```
ans = struct with fields:
        RiseTime: 7.2506
    SettlingTime: 121.6833
     SettlingMin: 47.8410
     SettlingMax: 54.9795
       Overshoot: 9.9590
      Undershoot: 0
            Peak: 54.9795
        PeakTime: 116
```

```
finalW = mean(w(t>=6));
finalWred = mean(wRef(t>=6));
SSgain = finalW/finalWred
```

```
SSgain = 1.0000
```

```
% We had the following requirements defined in 2. :
% 1 - Overshoot less than 10%
% 2 - 5%-settling time less than 0.2 seconds
% 3 - Steady state gain of 1±5%
```

```
% We see an OS% of 9.95 < 10 so OS requirements are met
% 5% settling time:  0.21 > 0.2 but as we are approxiamting
% this small error could be tolerated so Ts is satisfied and
% the SS gain is close to 1 and within our 5% error bound.

% So all the requirements are satisfied based on the plot above
```